

# UPENDRA GUPTA

C Developer · Linux · 5G RAN Engineer

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## SUMMARY

**3+ years** of C and Linux development in **5G RAN** at **Capgemini** — integrating **CBRS and NR-POS** features with **100% defect-free rollout** across US and European customers. Built automation tools that cut manual effort by **90%**. Awarded **3 times** by Capgemini for delivery excellence and innovation.

## EDUCATION

**B.E. Computer Science & Engineering** · Visvesvaraya Technological University, Bangalore

8.0/10 CGPA · 2022

## TECHNICAL SKILLS

**Languages:** C, Bash / Shell Scripting

**RAN / Telecom:** 5G NR, 3GPP, O-RAN, CBRS, NR-POS, gNB, CU-DU Split, Layer 2, SA, TDD, O1 Interface, NG Interface, F1 Interface, Call Flow

**Systems / Tools:** Linux, Git, Unit Testing, Multithreading, Core/Thread Affinity, NUMA, Memory Management, GDB, Wireshark, Valgrind, Core Dump Analysis, System Bring-up, Hugepages, Jenkins, CI/CD, Make, CMake

## WORK EXPERIENCE

**C Developer · Capgemini (formerly Altran), Gurugram**

Aug 2022 – Present

- **Integrated CBRS and NR-POS** modules into the 5G RAN stack in C, collaborating with global teams across US and Europe — maintained 3GPP/O-RAN compliance throughout.
- **Diagnosed CU-DU Split protocol issues** using GDB and Wireshark across Layer 2 boundaries, reducing recurring defects by ~40% and stabilising production deployments.
- **Achieved 90% defect-free rollout** of CBRS O1 feature across US and European customer environments through end-to-end integration and automated CFG validation.
- **Built a zero-touch server bring-up and E2E test framework** in C & Bash, eliminating 90% of manual setup and validation effort — adopted team-wide. Won Excellence Award 2024.
- **Developed internal CI/CD pipeline automation** on Linux, boosting developer productivity by 30% and removing manual intervention from the build-to-release cycle.

## PROJECTS

**CBRS O1 & NR Positioning Integration** | C · 3GPP · O1 Interface · CBRS · NR-POS · 5G SA

- **CBRS:** Integrated Citizens Broadband Radio Service and O1 management interface in C with full client-side validation — aligned with FCC Part 96 and 3GPP O-RAN specs. Zero defects at go-live across 3 US operator environments.
- **NR-POS:** Integrated 5G NR Positioning into the RAN stack, handling NG/F1 interface signalling and location service configuration. Deployed at multiple customer sites — 100% uptime in first 30 days post-launch.

**Zero-Touch Server Bring-Up & E2E Test Framework** | C · Bash · Linux

- **Built:** End-to-end automated pipeline covering OS provisioning, NUMA/core assignment, gNB bring-up, and functional validation — zero manual steps. 90% reduction in effort; framework adopted team wide.

## CERTIFICATIONS

- C for Everyone by [Coursera](#) (2022)
- 5G-architecture-design-protocols-evolution-deployment by [Udemy](#) (2024)
- 5G for Everyone by [Coursera](#) (2023)
- Linux Fundamental by [Coursera](#) (2023)
- Continuous Integration/Delivery by [Udemy](#) (2024)
- Jenkins by [Udemy](#) (2024)
- Ansible & Ansible-Playbooks for Automation by [Udemy](#) (2024)

## AWARDS

**WOW Award 2025** (CBRS/NR-POS Rollout) Recognized for deployments & integration of CBRS/NR-POS feature in Customer Setup.

**Excellence Award 2024** (Automation Framework) Recognized for developing C tools for zero-touch server bring-up and E2E test framework.

**WOW Award 2023** (Client Delivery) Recognized for exceptional client relationship management and consistently exceeding service delivery expectations.